I need 20 final-year project ideas for a Bachelor's in AI & Data Science,

due in 4 weeks (I have ~50-60 hours total to spend, alongside other commitments).

CONTEXT:

- Coder skill level: intermediate, will rely heavily on AI-assisted/"vibe" coding
- Must genuinely use AI/DS techniques (ML/DL/NLP/CV) — not just an API wrapper
- Should feel like a SaaS product, not just a script or notebook
- Available stack: Python, TensorFlow, scikit-learn, pandas, numpy, vector databases, LangChain/LangGraph, n8n, any frontend
- Deliverables needed: (1) a working demo app, (2) a report with real model evaluation (metrics, comparisons, not just "it works")

CONSTRAINTS — reject ideas that:

- Depend on data I can't realistically get in week 1 (scraping-heavy, proprietary, needs manual labeling of 1000s of samples)
- Are essentially solved/saturated in the market already (e.g. generic chatbot, generic chess engine, generic resume screener)
- Need real-time infra, telephony, or hardware I don't have access to
- Have legal/ethical landmines (medical diagnosis claims, unconsented call recording, financial advice without disclaimers)

FOR EACH OF THE 20 IDEAS, GIVE ME:

1. One-line pitch
2. The specific real-world problem it solves and for whom
3. What's genuinely novel/underserved about it vs existing tools
4. The core ML/DL technique involved (be specific: architecture, not just "AI")
5. A named public dataset it could realistically use (or state "needs synthetic/self-collected data" honestly if none exists)
6. Rough build risk: Low / Medium / High

Bias toward ideas that are unusual, regionally relevant, or solve a problem you don't see AI startups already doing to death. Avoid the most obvious/overdone categories unless you have a genuinely fresh angle on them.

Output as a numbered list, ideas only — no intro or summary text.Deep Research

All datasets verified. Now delivering the final 20 ideas.

1. Crop leaf disease triage app for smallholder farmers
2. Insect pest identification assistant for agriculture extension workers
3. Roman Urdu spelling normalizer for Pakistani SME customer feedback
4. Spelling-aware Roman Urdu review intelligence dashboard
5. Roman Urdu fake review detector for e-commerce platforms
6. Urdu fake news triage tool for newsroom editors
7. Claim veracity triage system for fact-checkers
8. Audio deepfake detector for media authenticity verification
9. Pakistan Sign Language gesture recognizer for accessibility
10. Urdu ASR correction dashboard for content creators
11. Flood building damage assessor for disaster response teams
12. Road damage reporter for municipal infrastructure monitoring
13. Smart parking occupancy monitor for facility managers
14. Dengue case-trend surveillance dashboard for public health researchers
15. Student dropout early-warning system for university administrators
16. Short-answer auto-grader with feedback for educators
17. Food calorie estimator for nutrition tracking
18. Smart-meter anomaly triage tool for utility operators
19. Bitcoin transaction fraud graph detector for compliance analysts
20. Multilingual waste sorter for recycling operations

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